

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination -Nov/ Dec -2024

ARTIFICIAL INTELLIGENCE [NPT-4014]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare the Turing Test with the Winograd Schema Challenge as methods of evaluating machine intelligence. What are the strengths and weaknesses of each? (5)
 - A)
 - B) Analyze the effectiveness of depth-first search versus breadth-first search in a specific problem context. (5)
- 2) Evaluate Hill Climbing as a heuristic search strategy. How does it perform in terms of efficiency and susceptibility to local optima? (5)
 - A)
 - B) Analyze the effectiveness of Genetic Algorithms in solving optimization problems, focusing on how selection, crossover, and mutation contribute to solution diversity. (5)
- 3) Evaluate the importance of heuristic admissibility in A* search and discuss how it impacts pathfinding optimality. (5)
 - A)
 - B) Compare Weighted A* and IDA* algorithms, focusing on their differences in memory usage and accuracy of solutions. (5)
- 4) Analyze the SSS* (Starred Minimax) algorithm's significance in game playing. Evaluate how it compares to traditional Minimax in terms of computational efficiency. (5)
 - A)
 - B) Evaluate the advantages and limitations of Goal Stack Planning in problem-solving. In what types of problems does this technique work best? (5)
- 5) Evaluate the AO* algorithm's suitability for solving AND-OR graphs. How effective is it in real-world applications that require problem decomposition? (5)
 - A)
 - B) Explain the function of a Rete Net in rule-based expert systems and discuss how it optimizes rule matching in large systems. (5)

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